COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

8 Further Human–Computer Interaction (lec40)

The design of conversational agents such as Echo or Siri has become very conventional with an activation phrase (e.g. 'hey assistant') and an action phrase (e.g. 'play encouraging music'). You're acting as the HCI researcher for a startup intending to disrupt the market by producing a radically different interaction design for the assistant you're developing.

Lecture 8: conversational agents

(*a*) What is a usability problem with conversational agents? [1 mark]

Answer: One of: invisible state, poor error correction, privacy

Full marks for any reasonable human centered critique.

Lecture 1: convergent and divergent design

(*b*) Why is iterative design important? What are the convergent and divergent phases of a design process? [3 marks]

Answer: Iterative design is important as the first idea isn't usually the best one. Iteration allows the incorporation of critique into a design.

The convergent design phase involves combining multiple designs into a smaller number of better ones, often based on a critical view of the designs.

The divergent design phase involves generating new design possibilities from existing ones.

Lecture 1: convergent and divergent design

(*c*) How would you go about performing the divergent phases on this project? What would you expect the output to be? [4 marks]

Answer: In order to perform the divergent phases, I would consider ways of swapping the communication modality from voice to other kinds of input. For example asking "how can we replace voice input with written input?" and "how can we replace voice input with physical input?"

The output would be a range of novel candidate designs for the interaction method, for example writing instructions on post-it notes.

3 marks for any reasonable idea generating strategy starting from existing designs to produce new ones, including ones that are not immediately practical. 1 mark for output being a range of designs.

Lecture 1: convergent and divergent design

(*d*) How would you go about performing the convergent phases on this project? What would you expect the output to be? [4 marks]

Answer: I would perform the convergent design phase by critiquing the proposed design using a theoretical frame. This would identify parts of the candidate design that have good usability properties and parts that don't. I would then combine the parts that have good usability together to create a smaller set of better design. An example of a relevant theoretical frame would be the attention investment framework.

This would create a smaller set of better designs.

Full marks for any reasonable combination strategy that attempts to use the best parts of existing range of designs.

Lecture 1: convergent and divergent design

(*e*) After following this process you have a candidate design. What are two properties that the design should have? [2 marks]

Answer: Good feedback, visible system state

Full marks for any reasonable properties

Lecture 1: using theory in the design process, Lecture 7: evaluating interactive systems

(*f*) For *each* property, (i) name the underlying theory, (ii) what approach you would take to testing whether the design has that property, and (iii) a limitation of your method. [6 marks]

Answer: Theory: Cognitive Walkthrough / Theory of Exploratory Learning Method: Using an expert group of reviewers consider a set of goals and the actions that the user might take. Determine what the system would do, and whether that would help the user achieve their goal. Limitation: Death by detail, too much feedback to usefully use to improve the design

Theory: Cognitive Dimensions of Notations / Hidden Dependencies Method: Consider an activity such as exploration, using the interface determined by the design processes above, consider whether the next action the system will take is dependent on information that can't be seen on the interface Limtiation: Doesn't take into account whether the user can understand the information presented, which is likely to be a key limitation due to Role Expressiveness of information in systems making use of machine learning

Full marks for any reasonable selection of theory, method and limitation